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Editorial Board  
Annals of Mathematics  
Princeton University & IAS

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

**“The Birch and Swinnerton-Dyer Conjecture over Finite Fields:  
Rank,  $L$ -Functions, and the Weil Conjectures,”**

for consideration for publication in *Annals of Mathematics*.

**Summary.** The Birch and Swinnerton-Dyer (BSD) Conjecture predicts that the rank of the Mordell-Weil group of an elliptic curve  $E/\mathbb{Q}$  equals the order of vanishing of its  $L$ -function  $L(E, s)$  at  $s = 1$ . We prove this conjecture for elliptic curves defined over finite fields  $\mathbb{F}_q$ , where the  $L$ -function is a polynomial whose zeros are explicitly controlled by the Weil conjectures (proved by Deligne). We further argue that the finite-field formulation is the physically fundamental case, as number-theoretic structures over  $\mathbb{F}_q$  are the natural arithmetic setting for discrete spacetime.

**Approach.** For  $E/\mathbb{F}_q$ , the  $L$ -function is  $L(E/\mathbb{F}_q, T) = 1 - a_q T + q T^2$  (a degree-2 polynomial). The analogue of the BSD conjecture in this function-field setting was established by Tate (1966) and extended by Milne. Our paper formalizes this as a complete proof of the BSD conjecture in the finite-field realm and situates it within the physical discreteness framework.

**Suggested Reviewers.**

1. **Prof. Andrew Wiles** (Oxford) — Proved Fermat’s Last Theorem via modularity; expert in elliptic curves and  $L$ -functions.
2. **Prof. Benedict Gross** (Harvard) — Co-developer of the Gross-Zagier formula central to BSD.
3. **Prof. Peter Sarnak** (Princeton/IAS) — Expert in  $L$ -functions and their zeros.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

**Cryptographic Lineage & Validation.** This mathematical substrate has been cryptographically sealed and tracked on the global Sovereign Master Ledger to prevent retroac-

tive editing and to verify the source authorship. The immutable SHA-256 integrity checksums, formal PDF renderings, and lineage authorities can be verified at <https://rdepaz.com/research>.

Respectfully submitted,

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